

Run this script to verify your environment is working correctly:

```
1. python test_environment.py
```

You should see the library versions, some Apple stock data, and a success message.

Maintaining Your Python Environment

Setting up your environment is just the beginning. Maintaining it properly is crucial for long-term success in algorithmic trading.

Regular Updates: Security and Performance

Regularly updating your libraries is important for both security and performance. However, in a trading context, you need to be careful about updates potentially changing the behavior of your algorithms.

In November 2021, a major cryptocurrency trading firm lost approximately \$1.2 million when a seemingly minor update to a mathematical library subtly changed how floating-point calculations were rounded in edge cases. This affected their arbitrage algorithms, which relied on precise calculations of price differences.

To update your libraries safely:

1.

Create a separate test environment before updating production code:

```
1. conda create --name test_env --clone trading_env
2. conda activate test_env
3. pip install --upgrade -r requirements.txt
4.
```

2.

Run your backtests in the test environment to verify performance hasn't changed

3.

Only then update your main environment

Environment Documentation

Document your environment thoroughly. This is standard practice at firms like Citadel and Two Sigma, where reproducibility is critical.